

Problem Set 1

Hoa T. Vu

Gradient Descent

Problem 1: Mean Squared Error. Given the curve $\hat{y} = 4x - 2$ and the data below, compute the MSE.

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

x	1	2	3	4
y	3	7	9	15

Problem 2: Compute the Gradient. Find ∇f for each function.

(a) $f(x, y) = x^2y + e^x$

(b) $f(x, y, z) = x \ln(y) + yz^2$

(c) $f(x, y) = \sin(xy) + x^3$

(d) $f(x, y) = \frac{x^2}{y+1}$

Problem 3: Do These Schedules Converge?

Recall: a learning-rate schedule (α_t) satisfies the Robbins–Monro conditions if

$$\sum_{t=1}^{\infty} \alpha_t = \infty \quad \text{and} \quad \sum_{t=1}^{\infty} \alpha_t^2 < \infty.$$

Tool. The p -series $\sum_{t=1}^{\infty} \frac{1}{t^p}$ converges if and only if $p > 1$.

For each schedule, determine whether both conditions hold.

(a) $\alpha_t = \frac{5}{t+1}$

(b) $\alpha_t = e^{-t}$

(c) $\alpha_t = \frac{1}{\sqrt{t+1}}$

Python Programming

Problem 4: Lists. Given the list below, write Python code for each task.

```
scores = [88, 72, 95, 61, 84, 90]
```

- (a) Access the third score.
- (b) Slice the last three scores.
- (c) Append 77, then sort in descending order.
- (d) Remove all scores below 80 using a list comprehension.

Problem 5: Sieve of Eratosthenes. Find all primes up to n by repeatedly crossing out multiples:

1. Start with `is_prime = [True] * (n + 1)`; set `is_prime[0] = is_prime[1] = False`.
2. For each p from 2 to $\lfloor \sqrt{n} \rfloor$: if `is_prime[p]`, mark `is_prime[p*p]`, `is_prime[p*p + p]`, ... as `False`.
3. Collect all indices where `is_prime[i]` is `True`.

Tasks:

- (a) Implement `sieve(n)` and verify it returns `[2, 3, 5, 7, 11, 13]` for $n = 13$.
- (b) Use it to find the **1000th prime**.
- (c) Count how many primes are below 10^6 . What is the running time of the sieve?